

UNIT-I: Understanding Cloud Computing

Textbook-aligned & university-exam-oriented student presentation

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Cloud Computing: Concepts, Technology & Architecture

Unit-I Learning Focus

Understand what cloud computing means.

Identify important characteristics of cloud computing.

Understand the evolution of cloud computing.

Recognize common cloud computing use cases.

Explain major benefits and challenges of cloud adoption.

1. What is Cloud Computing?

Cloud computing is a way of obtaining IT resources and services remotely through a cloud environment.

The book emphasizes cloud consumers, cloud providers, cloud-based IT resources and cloud services.

Instead of focusing only on physical computers, cloud computing focuses on remotely accessible and provisioned IT resources.

Key idea: computing capabilities can be consumed as services.

Cloud Provider and Cloud Consumer

Cloud Provider: the party that makes cloud services or IT resources available.

Cloud Consumer: the party that uses the cloud service or cloud-based IT resource.

A cloud environment establishes a relationship between provider and consumer.

This provider–consumer view is important throughout the book.

Simple Example of Cloud Computing

Traditional approach: an organization purchases and manages its own servers.

Cloud approach: the organization obtains computing resources from a cloud provider when required.

Example uses: application hosting, storage, backup, development and testing.

Key idea: consume IT capabilities as remotely accessible services.

2. Evolution of Cloud Computing

Cloud computing did not appear suddenly; it developed from earlier computing and networking approaches.

The book discusses the history of cloud computing together with business drivers and technology innovations.

Organizations increasingly wanted flexible, scalable and remotely accessible IT capabilities.

Cloud computing brought these capabilities together into service-oriented computing environments.

Evolution – Simple View

Traditional / centralized computing



Networked and distributed computing



Virtualized infrastructure and data centers



Web-based services and service-oriented approaches



Cloud computing environments

Why Cloud Computing Emerged

Growing amounts of data and increasing computing requirements.

Need for flexible and scalable IT capabilities.

Need for remotely accessible IT resources.

Business drivers encouraged organizations to consume IT as services.

Technology innovations helped make cloud-based service delivery practical.

3. Characteristics of Cloud Computing

The book's Chapter 4 identifies six common characteristics:

1. On-demand usage
2. Ubiquitous access
3. Multitenancy and resource pooling
4. Elasticity
5. Measured usage
6. Resiliency

Characteristic 1 – On-Demand Usage

A cloud consumer can access cloud-based IT resources when required.

The consumer can self-provision resources.

After configuration, usage can be automated with little or no further human involvement.

Example: creating a cloud resource when workload increases.

Characteristic 2 – Ubiquitous Access

Cloud services are designed to be widely accessible.

Access may involve different devices, protocols, interfaces and security technologies.

The service architecture may support different types of cloud consumers.

Simple idea: access the service wherever an appropriate network connection is available.

Characteristic 3 – Multitenancy & Resource Pooling

Multitenancy allows one software instance or environment to serve multiple consumers (tenants).

Tenants are logically isolated from one another.

Resource pooling allows physical and virtual resources to be dynamically assigned and reassigned.

Virtualization is commonly used to support resource pooling.

Characteristic 4 – Elasticity

Elasticity is the ability of IT resources to adapt to changing workload demands.

Resources can be increased when demand rises.

Resources can be reduced when demand falls.

Elasticity supports efficient use of resources.

Characteristic 5 – Measured Usage

Cloud environments can measure resource/service usage.

Measurement supports usage-based monitoring and management.

It also supports usage-related charging and analysis.

Think: 'How much resource did the consumer use?'

Characteristic 6 – Resiliency

Resiliency is the ability of IT resources to recover from operational disturbances.

It can involve redundancy, replication and recovery mechanisms.

Important textbook note: NIST's definition lists five characteristics; the book additionally includes resiliency as a common cloud characteristic.

Characteristics – Quick Comparison

On-demand usage → obtain resources when needed.

Ubiquitous access → broadly accessible services.

Multitenancy/resource pooling → shared and dynamically assigned resources.

Elasticity → resources adapt to workload.

Measured usage → usage can be monitored and measured.

Resiliency → recover from operational disturbances.

4. Cloud Computing Use Cases

Cloud computing can be used wherever remotely accessible IT resources or services provide value.

Online application hosting.

Cloud storage and backup.

Development and testing environments.

Data processing and analytics.

Web and business applications.

Use Cases Connected to Characteristics

Application hosting → on-demand and scalable resources.

Cloud storage → ubiquitous access and resource pooling.

Development/testing → flexible provisioning.

Analytics → access to scalable computing resources.

Business applications → remotely accessible services.

The appropriate use case depends on the requirements of the organization.

5. Benefits of Cloud Computing

Access to IT resources without necessarily owning all underlying infrastructure.

Flexible resource provisioning.

Scalability and elasticity for changing workloads.

Resource pooling and better utilization.

Measured usage supports monitoring and usage-based management.

Organizations can focus more on services and applications rather than all underlying infrastructure.

6. Challenges of Cloud Computing

Dependence on network connectivity and remote access.

Security and trust considerations.

Reliability and availability concerns.

Loss of direct control over underlying infrastructure.

Migration, interoperability and portability considerations.

Cost management can become important as cloud usage grows.

Benefits vs Challenges

Benefit: flexible provisioning ↔ Challenge: cost management.

Benefit: remote access ↔ Challenge: network dependence.

Benefit: shared resources ↔ Challenge: security and trust.

Benefit: provider-managed infrastructure ↔ Challenge: less direct control.

Benefit: scalable resources ↔ Challenge: migration, interoperability and portability.

Cloud Computing – Exam-Oriented Definition

Cloud computing is a way of obtaining IT resources and services remotely through a cloud environment.

Key terms to remember:

Cloud provider → makes resources/services available.

Cloud consumer → uses resources/services.

IT resources/services → computing capabilities consumed remotely.

Important University Examination Questions

Define cloud computing and explain its important characteristics.

Explain the evolution of cloud computing.

Explain the role of cloud provider and cloud consumer.

Explain the six characteristics of cloud computing.

Explain cloud computing use cases with suitable examples.

Explain the benefits of cloud computing.

Discuss the challenges of cloud computing.

Explain the evolution, benefits and challenges of cloud computing.

Unit-I Exam Priority

VERY HIGH: Definition of cloud computing, characteristics, evolution, benefits and challenges.

MUST UNDERSTAND: provider vs consumer, on-demand usage, ubiquitous access, multitenancy/resource pooling, elasticity, measured usage and resiliency.

MUST REMEMBER: suitable examples of cloud use cases and clear differences between benefits and challenges.

One-Page Revision

Cloud computing → remotely consumed IT resources/services.

Provider → makes resources/services available.

Consumer → uses resources/services.

Six characteristics → On-demand, Ubiquitous access, Multitenancy/resource pooling, Elasticity, Measured usage, Resiliency.

Use cases → Hosting, storage/backup, development/testing, analytics, web/business applications.

Benefits → Flexible access, provisioning, scalability, utilization, measured usage.

Challenges → Network dependence, security/trust, reliability, control, migration/portability, cost management.

Unit-I Final Recap

Cloud computing focuses on consuming IT capabilities as services.

Its evolution is linked to business needs and technology innovations.

Cloud environments provide important characteristics such as on-demand use, elasticity and measured usage.

Cloud computing offers flexibility and scalability but also introduces security, connectivity, reliability and management challenges.

These points form the core of Unit-I university examination preparation.